

DSC540WeekProjectMilestone3

May 7, 2025

1 Project Milestone 3

```
[165]: pip install html5lib
```

Requirement already satisfied: html5lib in /opt/anaconda3/lib/python3.12/site-packages (1.1)

Requirement already satisfied: six>=1.9 in /opt/anaconda3/lib/python3.12/site-packages (from html5lib) (1.16.0)

Requirement already satisfied: webencodings in /opt/anaconda3/lib/python3.12/site-packages (from html5lib) (0.5.1)

Note: you may need to restart the kernel to use updated packages.

```
[166]: # Importing Pandas packages required for this exercise.
```

```
import pandas as pd
import requests
import html5lib
import matplotlib
from matplotlib import pyplot as plt
```

```
[167]: # loading covid CSV file for data transformation and cleaning
```

```
url = 'https://en.wikipedia.org/wiki/COVID-19_pandemic_by_country_and_territory'
```

```
[169]: dfs = pd.read_html(url)
```

```
[172]: corona_df=dfs[12]
corona_df
```

```
[172]:
```

	Unnamed: 0 \
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
..	...
235	NaN
236	NaN

```

237                                     NaN
238                                     NaN
239 ^ Countries which do not report data for a col...

```

```

                                     Country \
0                                     World[a]
1                                     Peru
2                                     Bulgaria
3                                     North Macedonia
4                                     Bosnia and Herzegovina
..                                     ...
235                                    Pitcairn Islands
236                                    Tokelau
237                                    North Korea
238                                    Turkmenistan
239 ^ Countries which do not report data for a col...

```

```

                                     Deaths / million \
0                                     883
1                                     6601
2                                     5669
3                                     5422
4                                     5114
..                                     ...
235                                    0
236                                    0
237                                    0
238                                    0
239 ^ Countries which do not report data for a col...

```

```

                                     Deaths \
0                                     7057132
1                                     220975
2                                     38700
3                                     9978
4                                     16392
..                                     ...
235                                    0
236                                    0
237                                    0
238                                    0
239 ^ Countries which do not report data for a col...

```

```

                                     Cases
0                                     775866783
1                                     4526977
2                                     1329988

```

```

3          350924
4          403666
..          ...
235         4
236        80
237         0
238         0
239  ^ Countries which do not report data for a col...

```

[240 rows x 5 columns]

```
[173]: corona_df_col=corona_df.drop(columns=['Unnamed: 0'])
```

```
[177]: #Clean Country column data
```

```
corona_df_col['Country'].str.replace('[/,.*'],'')
```

```

[177]: 0          World[a]
1          Peru
2          Bulgaria
3          North Macedonia
4          Bosnia and Herzegovina
...
235         Pitcairn Islands
236         Tokelau
237         North Korea
238         Turkmenistan
239  ^ Countries which do not report data for a col...
Name: Country, Length: 240, dtype: object

```

```

[179]: # remove records where deaths are equals to 0 or less
corona_df_clean=corona_df_col[(corona_df_col['Deaths'])>"0"]

```

```
[181]: corona_df_clean
```

```

[181]:          Country \
0          World[a]
1          Peru
2          Bulgaria
3          North Macedonia
4          Bosnia and Herzegovina
..          ...
227         Tajikistan
228         Benin
229         Chad
230         Burundi
239  ^ Countries which do not report data for a col...

```

	Deaths / million \
0	883
1	6601
2	5669
3	5422
4	5114
..	...
227	12
228	11
229	10
230	1
239	^ Countries which do not report data for a col...

	Deaths \
0	7057132
1	220975
2	38700
3	9978
4	16392
..	...
227	125
228	163
229	194
230	15
239	^ Countries which do not report data for a col...

	Cases
0	775866783
1	4526977
2	1329988
3	350924
4	403666
..	...
227	17786
228	28036
229	7702
230	54569
239	^ Countries which do not report data for a col...

[232 rows x 4 columns]

```
[183]: # remove all the unwanted rows from the dataframe
```

```
corona_remove_un_rows=corona_df_col.drop([239])
```

```
[185]: corona_remove_un_rows
```

```
[185]:
```

	Country	Deaths / million	Deaths \
0	World[a]	883	7057132
1	Peru	6601	220975
2	Bulgaria	5669	38700
3	North Macedonia	5422	9978
4	Bosnia and Herzegovina	5114	16392
..	...	...	...
234	Saint Helena, Ascension and Tristan da Cunha	0	0
235	Pitcairn Islands	0	0
236	Tokelau	0	0
237	North Korea	0	0
238	Turkmenistan	0	0

```

Cases
0    775866783
1    4526977
2    1329988
3    350924
4    403666
..    ...
234    2166
235    4
236    80
237    0
238    0

```

[239 rows x 4 columns]

```
[234]: corona_df_1=dfs[15]
corona_df_1 = corona_df_1.rename(columns={'Region[30]': 'Region', 'Vaccinated_
↳%[31]': 'Vaccinated'})
corona_df_1 = corona_df_1[['Region', 'Cases per million']]
```

```
[236]: Covidsort = corona_remove_un_rows.sort_values(by = 'Deaths',ascending = True)
Covidsort
```

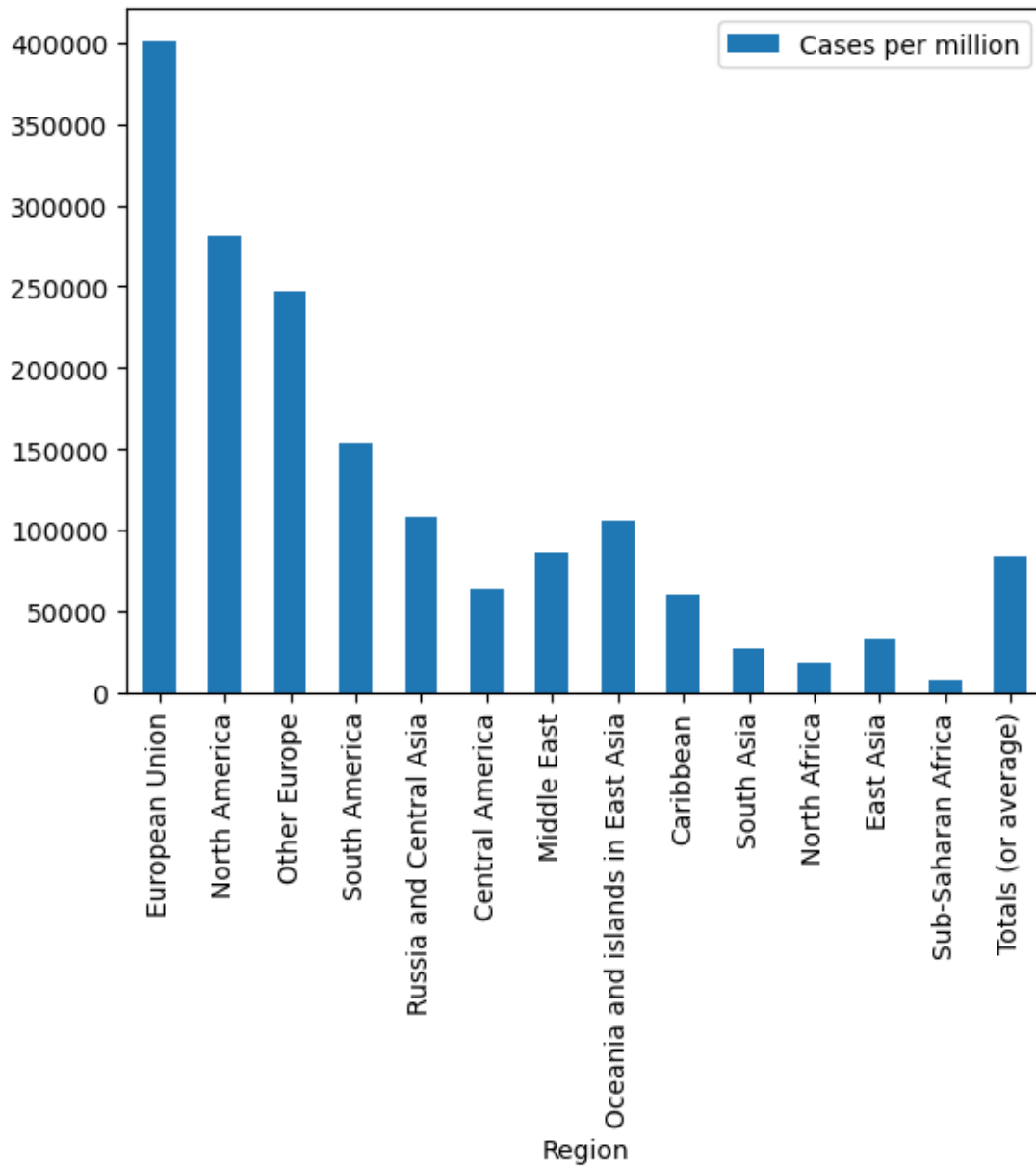
```
[236]:
```

	Country	Deaths / million	Deaths	Cases
238	Turkmenistan	0	0	0
232	Niue	0	0	1074
236	Tokelau	0	0	80
237	North Korea	0	0	0
235	Pitcairn Islands	0	0	4
..	...	...	...	...
76	Denmark	1642	9693	3435679
62	Republic of Ireland	1906	9744	1745088
18	Lithuania	3482	9810	1369355
167	Mauritania	204	997	63872

3 North Macedonia 5422 9978 350924

[239 rows x 4 columns]

```
[238]: corona_df_1.plot(x='Region', y='Cases per million', kind='bar')
plt.show()
```



```
[ ]:
```

```
[ ]:
```